

Lab 10.1 Machine Learning 3 - Real-time Sound Recognition for Classification

Utilizing MTConnect sound stream

Learning goals

Students will be able to:

1. Understand and utilize MTConnect sound data stream
2. Implement a CNN model to predict operational state of the vacuum pump
3. Build a real-time web-based monitoring system

1.1 Introduction

In this lab, we will implement real-time machine sound recognition using a CNN model and MTConnect sound stream to predict the operational state of the vacuum pump. This lab is broken down into three main sections: **1) Utilize MTConnect sound data stream, 2) Implement a CNN model you developed in the prelab using TensorFlow Lite, and 3) Build up the entire monitoring system.**

Note that all sample codes (almost complete) for this lab are available on [Github repo](#). If you want, please download them to Raspberry Pi.

To get started, Figure 1 illustrates a simplified sound recognition system. IIoT technologies and embedded machine learning models realize this system. The idea is that the Raspberry Pi listens to sound from the machine and recognize the operational state by AI model, and then we are able to monitor remotely the real-time state via a web-based dashboard.

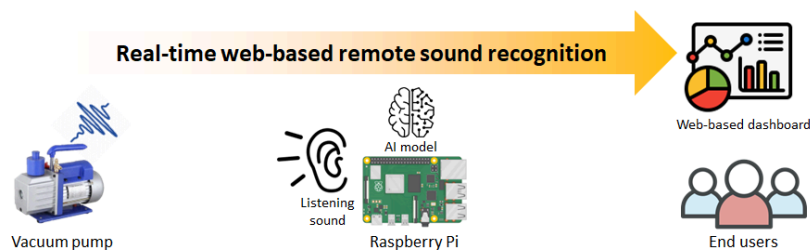


Figure 1 Simplified system configuration of vacuum pump sound recognition for Lab 10

Details of the system configuration and data flow are shown in Figure 2. Please take a close look at Figure 2. Let's say the MTConnect stream for sound is "sub-MTConnect" for sound to distinguish it from the MTConnect for results. First, the internal sound sensor connected to the headless computer captures analog sound signals and then converts them to digital signals. Second, the sub-MTConnect adapter samples, formats, and converts them to the sub-MTConnect agent. Therefore, as long as a client is accessible via TCP/IP to the headless computer, it can read the sound stream using the sub-MTConnect agent in real-time. Third, the edge computer (Raspberry Pi) gets the sound stream and takes a signal processing for the input feature of the CNN model that we developed in Prelab 10. For AI implementation, we utilize TensorFlow Lite on Raspberry Pi. Fourth, the outputs of the CNN model with the AI algorithm are taken to MTConnect. Fifth, the MTConnect data collector program aggregates MTConnect agent data and sends them to the MySQL database. Lastly, Grafana dashboard with the MySQL database can let end users know the current/history running state of the vacuum pump.

Target	Specification	Value	Unit/remark
	Middleware	MTConnect	
	Agent IP address	192.168.1.4	Router network (DNS: mepotrb16.ecn.purdue.edu)
	Port	5001	
	Data type	DiscreteTimeseries	
Sound stream	Data format	Signed 16-int	Full scale is -1 to 1.
	Sampling frequency	48	kHz
	Count/Sequence	2048	Approximately 43 msec
	dataItemId	sensor1	

Task 1.1

Using [curl](#) command below on Raspberry Pi and laptop, read current sound stream of the sensor and then attach it to the report as follows. Check the timestamp indicates current. It must be working on your laptop as well as long as both computers are on the same router network. Perform the same in the command prompt of your laptop as Raspberry Pi.

Raspberry Pi - Terminal AND Windows - Command Prompt

```
curl 192.168.1.4:5001/current
```

OR

```
curl mepotrb16.ecn.purdue.edu:5001/current
```

- Note that the headless computer is connected to the router network (DNS: mepotrb16.ecn.purdue.edu). Because of the port forwarding of 5001 to the headless computer (192.168.1.4), we can has the same results if you get access both IP address and DNS.

```

File Edit Tabs Help
pi@eunseobkkm:~
$ curl 192.168.1.4:5001/current
<?xml version='1.0' encoding='UTF-8'?>
<MTConnectStreams xmlns="urn:mtconnect.org:MTConnectStreams:1.4" xmlns:xsi="http://www.w3.org/2001/XMLSchema-Instance" xsi:schemaLocation="urn:mtconnect.org:MTConnectStreams:1.4 http://schemas.mtconnect.org/schemas/MTConnectStreams_1.4.xsd">
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(base) pi@robertcloud:~$ curl 10.165.67.146:5001/current
<?xml version="1.0" encoding="UTF-8"?>
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</MTConnectStreams>
(base) pi@robertcloud:~$
```



```
MINGW64/c:/Users/d908/OneDrive - purdue.edu/Graduate School/Spring 2026/ME-597-IIoT

r9D08Daniel-PC MTN6W64 - /OneDrive - purdue.edu/Graduate School/Spring 2026/ME-597-IIoT (main)
$ curl 10.165.67.146:5001/current
<?xml version="1.0" encoding="UTF-8"?>
<MTConnectStreams xmlns:ns="urn:mtconnect.org:MTConnectStreams:1.4" xmlns="urn:mtconnect.org:MTConnectStreams:1.4" xmlns: xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="urn:mtconnect.org:MTConnectStreams:1.4 http://schemas.mtconnect.org/XMLSchema-MTConnectStreams:1.4.xsd">
  <header creationTime="2026-04-10T14:24:20Z" sender="Iam-UltraPoint" instanceId="177568545" version="1.4.0.10" bufferSize="6192" nextSequence="3649706" firstSequence="3641514" lastSequence="3649705"/>
  <Streams>
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22 3664 3786 3885 3975 4002 4040 4044 4012 3969 3926 3843 3725 3585 3448 3285 3055 2812 2551 2265 1973 1695 1431 1162 884 599 313 39 -198 -437 -621 -776 -910 -1033 -1110 -
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1558 1664 1761 1895 2065 2201 2399 2439 2508 2561 2640 2744 2878 3009 3118 3211 3322 3411 3499 3562 3564 3545 3529 3465 3382 3311 3238 3120 3051 2976 2908 2852 2819 2752
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r9D08Daniel-PC MTN6W64 - /OneDrive - purdue.edu/Graduate School/Spring 2026/ME-597-IIoT (main)
$
```

1.3 Handling MTConnect sound stream in Python

Let's get access to the MTConnect sound stream and handle it using Python. Use [lab10_sample1.py](#) which is almost complete other than some constant variables. In the Python script, We will read approximately 1-second-long data at the moment for the CNN model we developed in the prelab. And then, signal processing and sound pressure level calculation will be made. Eventually, we will check the sound signal by plotting them.

The incomplete part of the sample script is below. You have to change the variables which is wrapped by aserteric mark (*). By completing the sample code , perform Task 1.2.

Python - Python3 (Part of [lab10_sample1.py](#))

```
## == CONSTANT ==
SAMPLE = "sample" # sample string for MTConnect HTML sample method
```

```

CURRENT = "current" # current string for MTConnect HTML current method
*SAMP_RATE* = int(?) # sampling rate
*CHUNK* = int(?) # chunk size
*AGENT* = "http://?ip?:?port?/" # MTConnect agent ip:port

*N* = int(?) # number of sequence to take sound
*N_FFT* = int(?) # number of FFT
*N_MELS* = int(?) # number of Mel filter bank

```

Task 1.2

To complete lab10_sample1.py, the required information is below. Note that you can use either Raspberry Pi or your laptop for this task.

- MTConnect (sound stream) agent IP: 192.168.1.4
- MTConnect (sound stream) agent port number: 5001
- Sampling frequency (sampling rate): MTConnect sound stream specification
- Chunk size: MTConnect sound stream specification
- Number of the required sequence: Same as CNN model training to make the sound signal approximately 1-second-long
- Number of FFT data points: Same as the CNN model training
- Number of Mel filter-bank: Same as the CNN model training

1. After completing the sample script, plot the signal as Figure 3 and then attach it on the report.
2. Calculate and print out the exact signal length in the second unit as Figure 4 and then attach the screenshot. You can use the commented-out line (line 57), but you have to replace "?" to get the answer.

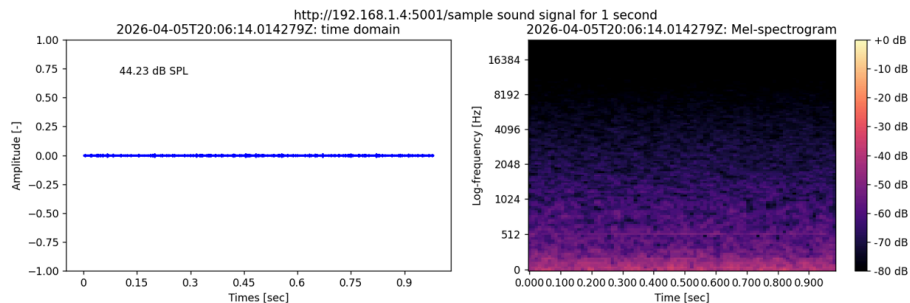


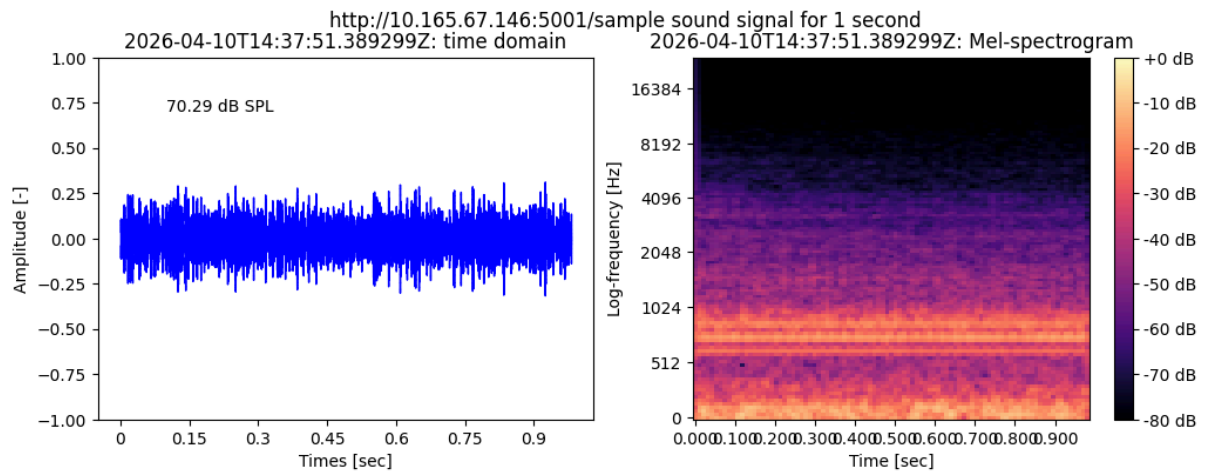
Figure 3 Sound signal plot: time domain (left) and Mel-spectrogram (right)

```

Current sequence info, timestamp: 2026-04-05T20:06:14.014279Z, firstSeq=19094, nextSeq=10902, lastSeq=19093
signal SPL (sound pressure level) is 44.23241963823554 dB.
Length of signal is 0.9813333333333333 [sec]
PS C:\Users\nobin>

```

Figure 4 Terminal after running lab10_sample1.py



```
Current sequence info, timestamp: 2026-04-10T14:37:51.389299Z, firstSeq=3667943, nextSeq=3659751, lastSeq=3667942
signal SPL (sound pressure level) is 70.29224457675441 dB.
Length of signal is 0.9813333333333333 [sec]
```

Please continue to [Lab 10.2 here](#).